In [1]:
```
import pandas as pd
from sklearn.linear_model import LinearRegression

df = pd.read_csv("ratings.tsv", sep="\t")
df.head()
```

Out[1]:
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>restaurant</th> <th>food</th> <th>ambience</th> <th>service</th> <th>rating</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody> </tbody>

1	85	82	89	78	
2	80	90	80	85	
3	83	86	83	85	
4	70	96	75	72	
5	68	80	78	75	

In [2]:
```
X = df[["ambience", "food", "service"]]
y = df["rating"]

model = LinearRegression()
model.fit(X, y)

print("Coefficients:")
for name, coef in zip(X.columns, model.coef_):
    print(name, ":", round(coef, 4))
```

```
Coefficients:
ambience : 0.1338
food : 0.7338
service : 0.324
```

In [3]:
```
coefficients = pd.DataFrame({
    "Factor": X.columns,
    "Coefficient": model.coef_
})

coefficients = coefficients.sort_values(
    by="Coefficient", ascending=False
)

print(coefficients.to_string(index=False))
```

```
   Factor  Coefficient
     food     0.733776
  service     0.323981
ambience     0.133761
```

I think the restaurant should improve its food first. Food has the highest coefficient (0.7338), so it has the strongest effect on the overall rating in this model. Service is second (0.3240), and ambience is last (0.1338).

In [ ]: